

Pillar 5: Capstone Project

Post Graduate Diploma in Artificial Intelligence and Machine Learning- September 2025

Heart-Disease-ML

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GitHub: <https://github.com/eschenry/Heart-Disease-ML.git>

ipynb file for all scripts:

Escalante Henry AIML Step 00 Pillar5-Capstone-ALLscripts.ipynb: for steps 02 to 05, separate scripts were prepared and uploaded into the GitHub

Original source of data:

url = "https://archive.ics.uci.edu/static/public/45/data.csv"

Load directly from github - contains the same data as data from the source

url = <https://raw.githubusercontent.com/eschenry/Heart-Disease-ML/refs/heads/main/data/data.csv>

Technical presentation file:

Business-facing presentation file: Escalante_Henry_Capstone-AI-and-ML-Business-facing-presentation-Heart-Disease-ML.pptx

Step-by-Step Instructions

Step 1: Problem Understanding & Framing

Step 1.1. Frame the business and data science problem clearly.

Step 1.1.a. Business Problem

Business Problem Statement

Healthcare providers often manage patients with diverse cardiovascular profiles. Understanding whether patients share similar clinical characteristics may help support population health studies, exploratory risk stratification, and future planning of monitoring strategies.

This study investigates whether patients can be grouped according to common cardiovascular characteristics such as age, blood pressure, cholesterol levels, heart rate, and other clinical indicators. Rather than focusing on prediction at this stage, the aim is to explore whether meaningful patient segments exist within the data.

If identifiable groups are present, they may provide useful insights for future research on preventive care programs, monitoring approaches, and resource allocation. However, the analysis is exploratory in nature and does not attempt to measure clinical outcomes, treatment effectiveness, or operational improvements.

The business objective is therefore to determine whether patients can be segmented into groups with broadly similar cardiovascular characteristics and to assess whether those groupings are meaningful enough to support further investigation.

Step 1.1.b. Data Science Problem

Data Science Problem Statement

From a data science perspective, the primary objective is to examine whether meaningful clustering structure exists within the Heart Disease dataset using unsupervised learning techniques.

The analysis is based on clinical variables including:

- Cholesterol
- Resting blood pressure
- Maximum heart rate
- ST depression (oldpeak)
- Number of major vessels (ca)
- Fasting blood sugar

The heart disease outcome variable is excluded during clustering so that any identified groups emerge from similarities within the patient characteristics themselves.

Rather than assuming that distinct patient groups already exist, the analysis evaluates whether the data exhibits sufficient clustering tendency to justify segmentation. Cluster quality is assessed using the Elbow Method, Silhouette Score, Dendrogram Analysis, and Hopkins Statistic.

Any resulting clusters should be viewed as approximate patient groupings intended for exploratory analysis rather than definitive cardiovascular risk categories.

Step 1.2. Define whether it's a classification, regression, recommendation, anomaly detection, or clustering task.

The primary task in this study is **clustering**, specifically **unsupervised hierarchical clustering**. The objective is to group patients based on similarities in their clinical characteristics without making use of the heart disease outcome variable.

In addition to clustering, supervised classification models were developed as a secondary analysis to examine how well the clinical variables predict heart disease status. Consequently, the project includes both clustering and classification techniques, although patient segmentation remains the main focus of the study.

Step 1.3. Specify success metrics (e.g., Accuracy, AUC, RMSE, Silhouette Score) and business KPIs (e.g., cost savings, uplift).

Success Metrics (Technical + Business)

Data Science Success Metrics

- **Hopkins Statistic:** To evaluate whether meaningful clustering tendency exists within the dataset; Hopkins Statistic > 0.5 indicates clustering tendency.
- **Silhouette Score:** To measure how well observations fit within their assigned clusters while remaining separate from other clusters. Silhouette Score > 0.35 indicates acceptable separation.
- **Elbow Method:** To identify an appropriate number of clusters; identifies a stable cluster count.
- **Dendrogram Analysis:** To visualize hierarchical cluster structure.
- **Cluster Interpretability:** To assess whether the resulting groups exhibit meaningful clinical characteristics; Clusters are clinically interpretable.

Business KPIs

- Identification of at least two patient segments with distinguishable clinical characteristics.
- Clear differences between cluster profiles.
- Cluster patterns that are consistent with established cardiovascular risk indicators.

- Evidence that the resulting groups may support exploratory patient stratification and further clinical investigation.

Step 2: Data Collection & Understanding

Using the Heart Disease Dataset (UCI) database.

Dataset Selection

This study uses the **UCI Heart Disease Dataset**, a well-known dataset that has been widely used in cardiovascular research and machine learning applications. The dataset contains a mix of demographic information, clinical measurements, and exercise-test results that make it suitable for both clustering and predictive modelling. Another advantage is that the dataset is publicly available, allowing results to be reproduced and compared with findings from previous studies.

Step 2.1. Summarize feature types, missing values, outliers, etc.

The dataset contains a mixture of numerical and categorical variables that describe different aspects of cardiovascular health.

Feature Types

- **Continuous numerical variables:** age, trestbps, chol, thalach, oldpeak
- **Binary categorical variables:** sex, fbs, exang
- **Categorical / ordinal variables:** cp, restecg, slope, thal
- **Discrete numerical variables:** ca
- **Target variable:**

- target (binary version (0 = no disease, 1 = disease)) OR num (original UCI version: 0, 1-4)

Data Quality Assessment

A review of the dataset showed that **no duplicate records** were present.

Duplicates: 0

Variables Containing Missing Values:

Variable	Missing Values
ca (number of major vessels)	4
thal (thalassemia test result)	2

Nulls (missing values):

age 0
sex 0
cp 0
trestbps 0
chol 0
fbs 0
restecg 0
thalach 0
exang 0
oldpeak 0
slope 0
ca 4
thal 2
num 0

All remaining variables were complete.

Outlier Assessment

Potential outliers were identified using the Interquartile Range (IQR) method.

Variable	Lower Bound	Upper Bound	Outliers Identified
Cholesterol (chol)	115	371	5
Resting Blood Pressure (trestbps)	90	170	9
ST Depression (oldpeak)	-2.40	4.00	5
Maximum Heart Rate (thalach)	84.75	214.75	1

Most outliers occurred in resting blood pressure, cholesterol, and ST depression, while maximum heart rate showed relatively few extreme values.

Distributional Characteristics

Histogram analysis revealed differences in the distributions of several variables:

- **Cholesterol** displayed a noticeable right-skew, with a small number of patients having unusually high values.
- **Oldpeak** was also positively skewed, indicating that higher levels of exercise-induced ST depression occurred less frequently.
- **Age** was distributed across a relatively broad range, with most patients falling between approximately 45 and 65 years.
- **Maximum heart rate (thalach)** showed substantial variation across patients but did not appear heavily skewed.

Overall, the dataset demonstrated reasonable data quality, with only a small number of missing values and a manageable number of outliers identified in a subset of variables.

Step 2.2. Provide a **data dictionary** (variables, types, units, allowed values).

data_dict

	Variable	Data Type	Missing Values	Unique Values
age	age	int64	0	41
sex	sex	int64	0	2
cp	cp	int64	0	4
trestbps	trestbps	int64	0	50
chol	chol	int64	0	152
fbs	fbs	int64	0	2
restecg	restecg	int64	0	3
thalach	thalach	int64	0	91
exang	exang	int64	0	2
oldpeak	oldpeak	float64	0	40
slope	slope	int64	0	3
ca	ca	float64	4	4
thal	thal	float64	2	3
num	num	int64	0	5

Data Dictionary (Variables, Types, Units, Allowed Values)

Variable	Description	Type	Unit	Allowed Values
	Patient Demographics			
Age	Age of patient	Continuous Numeric	Years	Typically 29-77
Sex	Gender	Binary Categorical	N/A	0 = Female, 1 = Male

	Chest Pain & Cardiac Indicators			
Cp	Chest pain type	Categorical	N/A	0 = Typical Angina, 1 = Atypical Angina, 2 = Non-anginal Pain, 3 = Asymptomatic
Trestbps	Resting blood pressure	Continuous Numeric	mm Hg	Positive numerical values; Typical observed range: 94–200 mmHg
Chol	Serum cholesterol	Continuous Numeric	mg/dL	Positive numerical values; Typical observed range: 126–564 mg/dL
Fbs	Fasting blood sugar > 120 mg/dL	Binary Categorical	N/A	0 = False, 1 = True
Restecg	Resting ECG result	Categorical	N/A	0 = Normal, 1 = ST-T Abnormality, 2 = Left Ventricular Hypertrophy
	Exercise & Stress Test Features			
thalach	Maximum heart rate achieved	Continuous Numeric	Beats per minute (bpm)	Positive numerical values; Typical observed range: 71–202 bpm
exang	Exercise-induced angina	Binary Categorical	N/A	0 = No, 1 = Yes
oldpeak	ST depression induced by exercise	Continuous Numeric	N/A	Typical observed range: 0.0–6.2
slope	Slope of peak exercise ST segment	Ordinal Categorical	N/A	0 = Upsloping, 1 = Flat, 2 = Downsloping
	Fluoroscopy & Thalassemia			

ca	Number of major vessels colored by fluoroscopy	Discrete Numeric	Count	0-4
thal	Thalassemia test result	Categorical	N/A	3 = Normal, 6 = Fixed Defect, 7 = Reversible Defect
	Target Variable			
target	Presence of heart disease	Target Variable	N/A	Original UCI: 0-4 disease severity scale. 0 = No disease, 1-4 = Disease present.

Step 3: Data Preprocessing, Applied EDA & Feature Engineering

- **Clean data:** Handle nulls, duplicates and outliers.
- **Engineer features:** Scaling, encoding, binning and domain-derived features.
- **Applied EDA:** Distributions, relationships, clustering tendency (if unsupervised).
- **Feature importance & explainability:** SHAP, LIME, or model-based importances.
- **Feature selection:** At least one approach (filter, wrapper, or embedded).
- **Dimensionality reduction:** PCA (and t-SNE/UMAP for visualisation if needed).

Step 3.1. Clean data: Handle nulls, duplicates and outliers.

The workflow included handling missing values, duplicate records, data type conversion and outlier treatment.

3.1.1 Initial Data Quality Assessment:

Initially checked were Duplicate Records and Missing Values Before Cleaning

No duplicate records were identified: Duplicates: 0:

3.1.2 Standardize Missing Values

The UCI Heart Disease dataset may encode missing values using the "?" character. These entries were first converted to standard missing values (NaN) before further processing.

```
df = df.replace("?", np.nan)
```

Variables Containing Missing Values:

Variable	Missing Values
ca (number of major vessels)	4
thal (thalassemia test result)	2

3.1.3 Convert Columns to Numeric (Convert Data Type)

After standardizing missing values, all columns were converted to numeric format where possible. Any invalid values were coerced to NaN.

```
df = df.apply(lambda col: pd.to_numeric(col, errors='coerce'))
```

3.1.4 Remove Duplicate Records

Duplicate observations were identified and removed to avoid repeated records influencing model training and evaluation.

```
print("Duplicates:", df.duplicated().sum())  
df = df.drop_duplicates()
```

There are NO duplicates

3.1.5 Missing Value Imputation

Missing values were imputed according to variable type:

- Median imputation for numerical variables;
- Mode imputation for categorical variables (ca, thal)

Median imputation was selected because it is less sensitive to extreme values than mean imputation.

3.1.6 Outlier Detection & Treatment (IQR Method)

Potential outliers were identified using the Interquartile Range (IQR) method and capped at the calculated upper and lower bounds.

```
num_cols = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak']  
for col in num_cols:
```

```

Q1 = df[col].quantile(0.25)
Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
df[col] = np.clip(df[col], lower, upper)

```

```

Outlier Treatment Summary
   Feature  Lower Bound  Upper Bound  Outliers Capped
0      age          28.50         80.50              0
1  trestbps          90.00        170.00              9
2      chol         115.00        371.00              5
3   thalach          84.75        214.75              1
4   oldpeak          -2.40           4.00              5

```

Variables treated for outliers included:

- Age (age)
- Resting blood pressure (trestbps)
- Cholesterol (chol)
- Maximum heart rate (thalach)
- ST depression (oldpeak)

Outlier capping was preferred over row deletion to preserve the limited sample size while reducing the influence of extreme observations.

Feature-by-Feature Interpretation

Age Interpretation

No age values fell outside the acceptable range.

Resting Blood Pressure (trestbps) Interpretation

Nine patients had resting blood pressure values lower than 90 mmHg or higher than 170 mmHg.

These values were capped to remain within the acceptable range. Resting blood pressure exhibited the largest number of extreme observations among the evaluated variables.

Cholesterol (chol) Interpretation

Five patients had cholesterol values outside the expected range.

This is not unusual because cholesterol measurements often contain a small number of unusually high values.

Maximum Heart Rate (thalach) Interpretation

Only one observation was considered extreme.

This indicates that the maximum heart rate values were generally consistent across patients.

ST Depression (oldpeak) Interpretation

Five observations exceeded the acceptable upper threshold of 4.0.

These values were capped since that small number of patients exhibited unusually high ST depression measurements relative to the rest of the sample.

Cleaning Completed Successfully

All records were cleaned using a consistent and reproducible workflow.

Missing values were standardized and imputed, duplicate observations were removed.

Outlier treatment was performed using the Interquartile Range (IQR) method. No outliers were identified for age, whereas resting blood pressure (trestbps) contained nine outlying observations, followed by cholesterol (chol) and ST depression (oldpeak) with five observations each. Maximum heart rate (thalach) contained one outlier.

The extreme values were capped at the calculated upper and lower bounds to reduce the influence of unusual observations while preserving the limited sample size.

Only a modest number of values required adjustment, indicating generally reasonable data quality within the selected numerical variables.

These preprocessing steps improved data quality and prepared the dataset for subsequent feature engineering, exploratory analysis, clustering and predictive modeling.

Summary of Preprocessing Decisions

Data Quality Issue	Method Applied	Justification
Missing values encoded as "?"	Converted to NaN	Standardized missing value representation
Inconsistent data types	Numeric conversion using <code>pd.to_numeric()</code>	Enabled statistical analysis and modeling
Duplicate records	Removed using <code>drop_duplicates()</code>	Prevented duplicated observations from biasing results
Missing numerical values	Median imputation	Robust against outliers
Missing categorical values	Mode imputation	Preserved the most common category
Extreme values	IQR-based capping	Reduced influence of outliers while retaining observations

Step 3.2. Engineer features: Scaling, encoding, binning and domain-derived features.

Justification:

- **Scaling** is crucial for PCA, clustering and distance-based methods.

`X_selected = X_scaled[selected_features]` where Selected features: ['thal', 'ca',
`scaler = StandardScaler()`

`X_scaled = X.copy()`

`X_scaled[num_cols] = scaler.fit_transform(X[num_cols])`

Without scaling, Cholesterol would dominate PCA simply because it has larger numeric values.

Standardization ensures: age, chol, oldpeak, thalach, contribute on a comparable scale.

Numerical features were standardized using **StandardScaler**, which transforms each variable to have a mean of zero and a standard deviation of one. Standardization was performed prior to PCA, clustering and distance-based modeling to prevent variables with larger numerical ranges from disproportionately influencing the analysis.

- **Binning age** reflects clinical age groups.

```
X_scaled['age_bin'] = pd.cut(
    df['age'],
    bins=[20, 40, 50, 60, 80],
    labels=['20-40', '40-50', '50-60', '60+'])
```

- **Risk index** combines 'trestbps', 'chol', 'thalach', 'oldpeak' into a domain-inspired composite.

```
X_scaled['risk_index'] = (
    0.3 * df['trestbps'] +
    0.3 * df['chol'] -
    0.4 * df['thalach'] +
    0.5 * df['oldpeak']
)
```

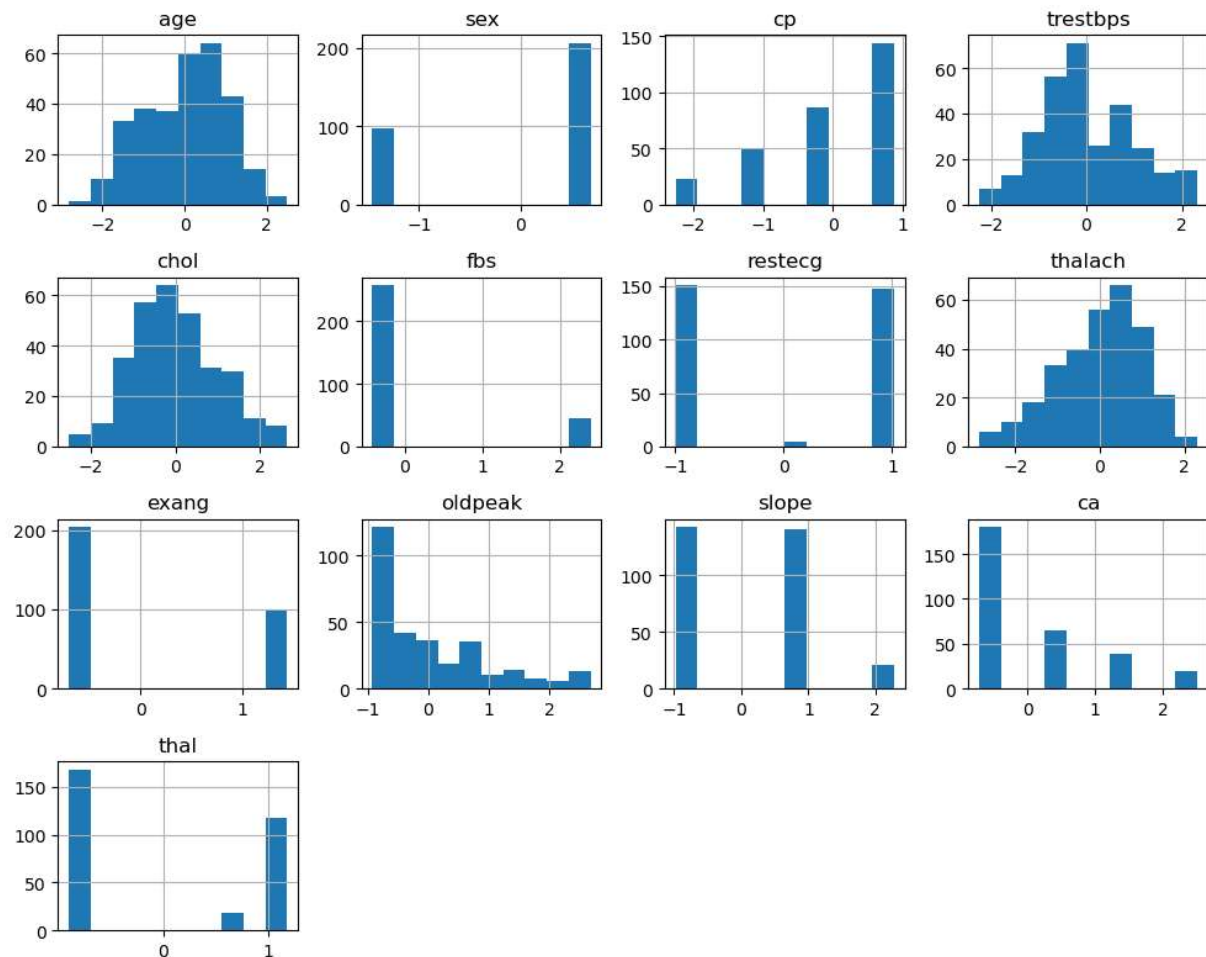
The risk_index feature was constructed as a domain-inspired composite measure combining resting blood pressure, cholesterol, maximum heart rate and ST depression. Positive weights were assigned to variables generally associated with increased cardiovascular risk, while a negative weight was assigned to maximum heart rate because lower exercise capacity is often linked to poorer cardiovascular health. The selected weights were heuristic and intended to represent relative risk contributions rather than clinically validated coefficients.

Step 3.3. Applied EDA: Distributions, relationships, clustering tendency (if unsupervised).

Interpretation:

- **Histograms** show skewness and ranges.
- **Pairplots/heatmaps** reveal relationships with target.
- **Hopkins statistic**: values closer to 0.5–0.7 suggest moderate clustering tendency; near 0 → strong structure, near 1 → random.

Step. 3.3.1. Distributions:

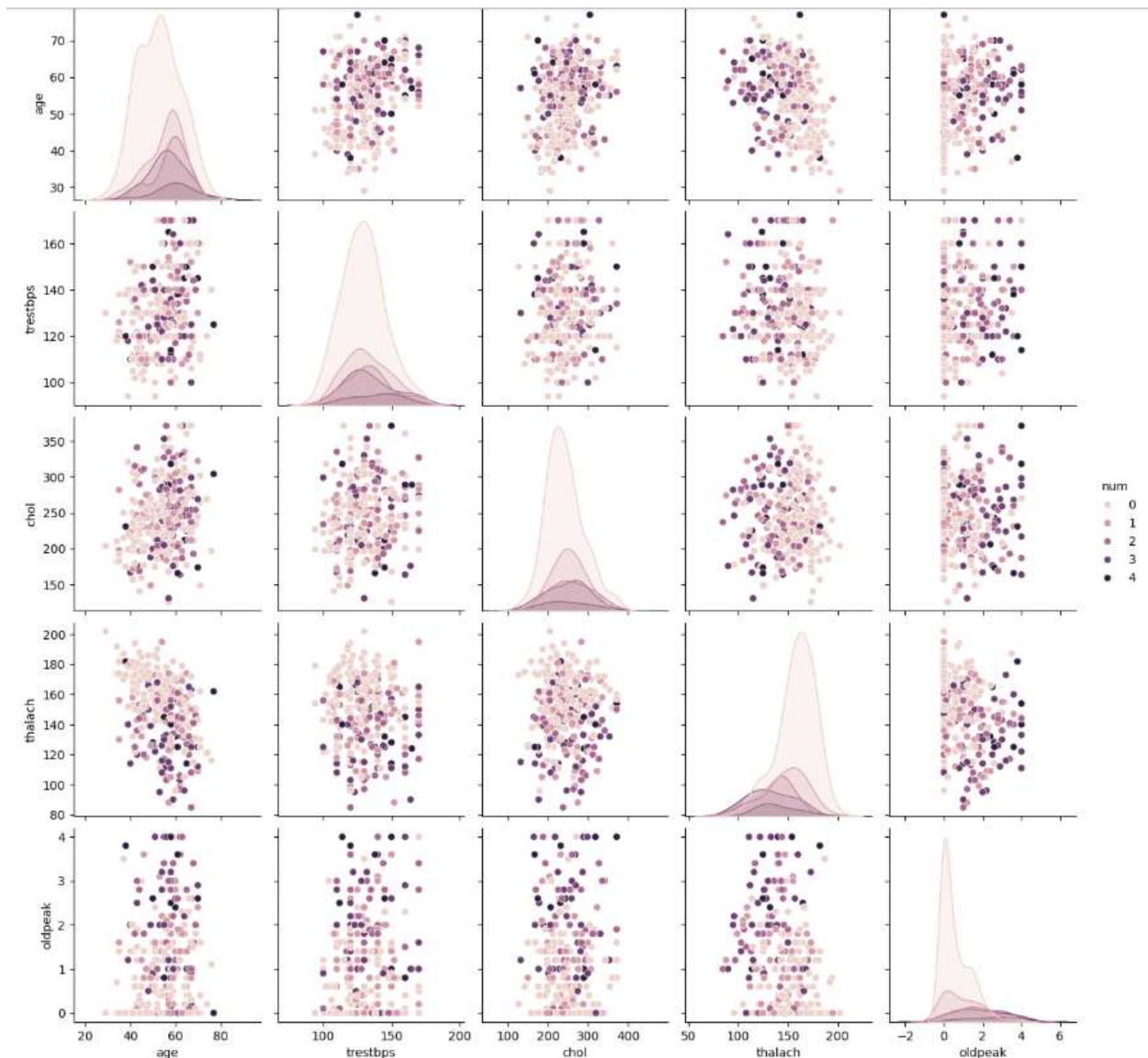


Step 3. Figure 1 - Histogram Distributions

A histogram shows: X-axis = feature values and that Y-axis = number of patients

Distribution plots were used to assess the range, central tendency and variability of the numerical variables. The histograms indicated that several features were not normally distributed and contained potential outliers. In particular, cholesterol and oldpeak appeared positively skewed, whereas age and maximum heart rate displayed relatively broader distributions. These observations informed subsequent preprocessing decisions, including scaling and outlier treatment. The distributions also suggest that cardiovascular risk factors vary substantially across patients within the dataset.

Step 3.3.2. Relationships (pairplot subset):



Step 3. Figure 2 – Pairplot Relationships

A pairplot contains two types of graphs:

1. Diagonal Plots: These are distributions (histograms or density plots). Examples: age, chol, oldpeak

2. Off-Diagonal Plots: These are scatterplots. They show: Whether two variables move together and whether the classes occupy different regions.

Examples: age vs chol, age vs oldpeak and chol vs thalach

Interpretation:

Pairplots were used to examine relationships among the main clinical variables and to assess whether patients with and without heart disease occupied different regions of the feature space.

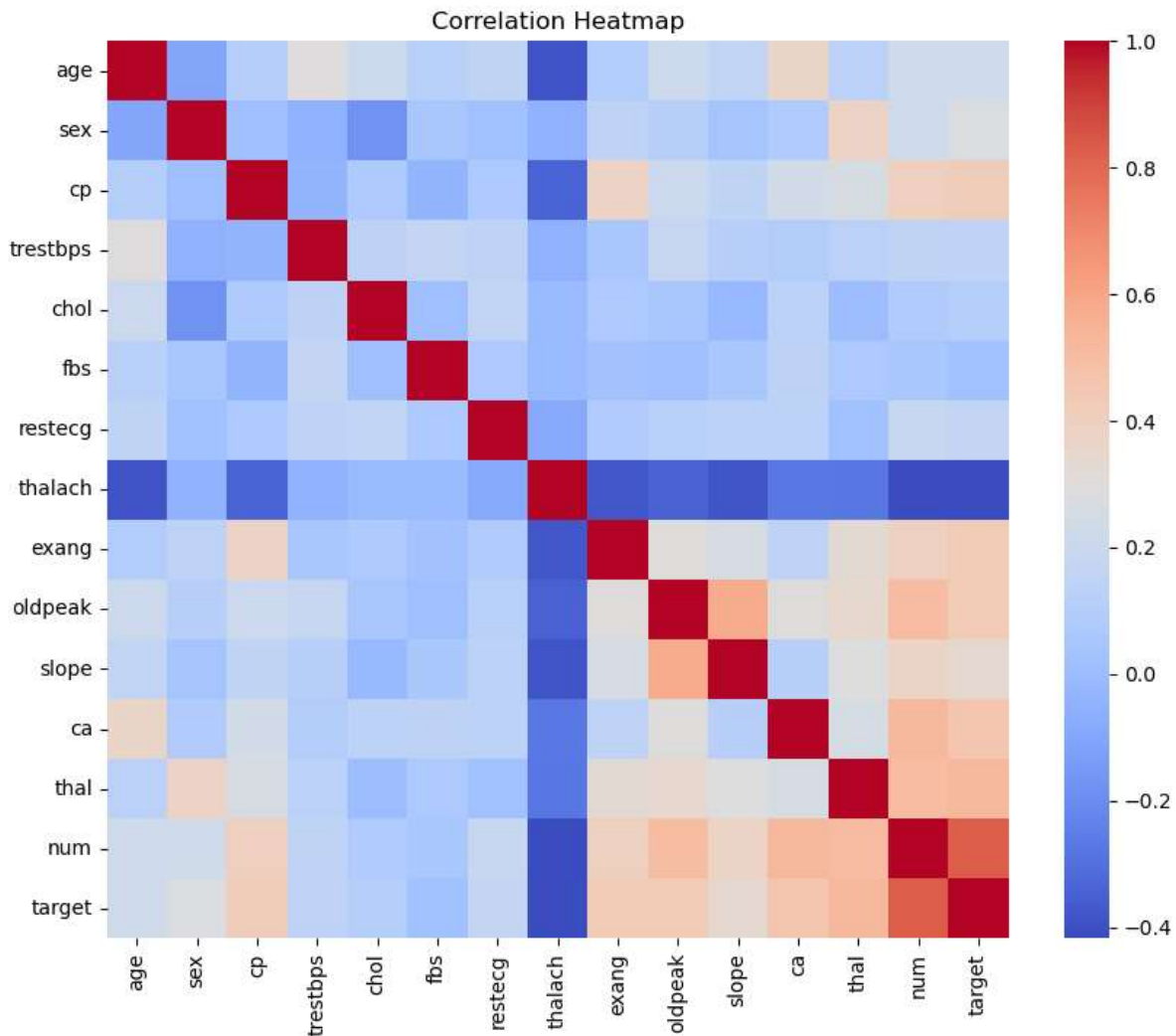
Several variables exhibited partial differences between classes; however, substantial overlap was observed across most feature combinations.

These findings suggest that no single variable pair provides clear separation between patients with and without heart disease. Nevertheless, the visualizations indicate potential multivariate relationships that may be exploited by machine learning models. The results are consistent with the strong performance of ensemble methods such as Random Forest, which can capture complex interactions among multiple predictors.

Key Takeaway

The pairplots show that heart disease status is not explained by any single variable alone. Instead, predictive information appears to emerge from the combined relationships among several clinical features, supporting the use of multivariate machine learning approaches.

Step 3.3.3. Correlation matrix



Step 3. Figure 3 – Correlation Heatmap

Correlation values range from: -1.0 to +1.0

Positive Correlation: example is for “age ↔ oldpeak = +0.45”

Interpretation: Patients with higher age tend to have higher oldpeak values.

Not always, but there is a general tendency.

Negative Correlation: example is “thalach ↔ target = -0.40”

Interpretation: Patients achieving higher maximum heart rates tend to be less likely to have heart disease.

No Correlation: example is “fbs ↔ target = 0.02”

Interpretation: There is little evidence of a linear relationship between fasting blood sugar and heart disease status.

Stronger Associations with Heart Disease: Variables such as: thal, ca, cp, oldpeak, exang show stronger correlations with target.

Weaker Associations: Variables such as: fbs, restecg show weaker connections.

Supports Feature Engineering Decisions

If several variables move together: chol, trestbps, oldpeak, it may justify creating: a risk_index, as a combined feature.

Most correlations were moderate in magnitude. Variables such as thal, ca, oldpeak, and chest pain type showed stronger associations with heart disease status, while fasting blood sugar and resting ECG exhibited weaker relationships.

This indicates that heart disease status is likely influenced by the combined effect of multiple factors rather than any single predictor alone.

Step 3.3.4. Clustering tendency (Hopkins statistic – simple implementation)

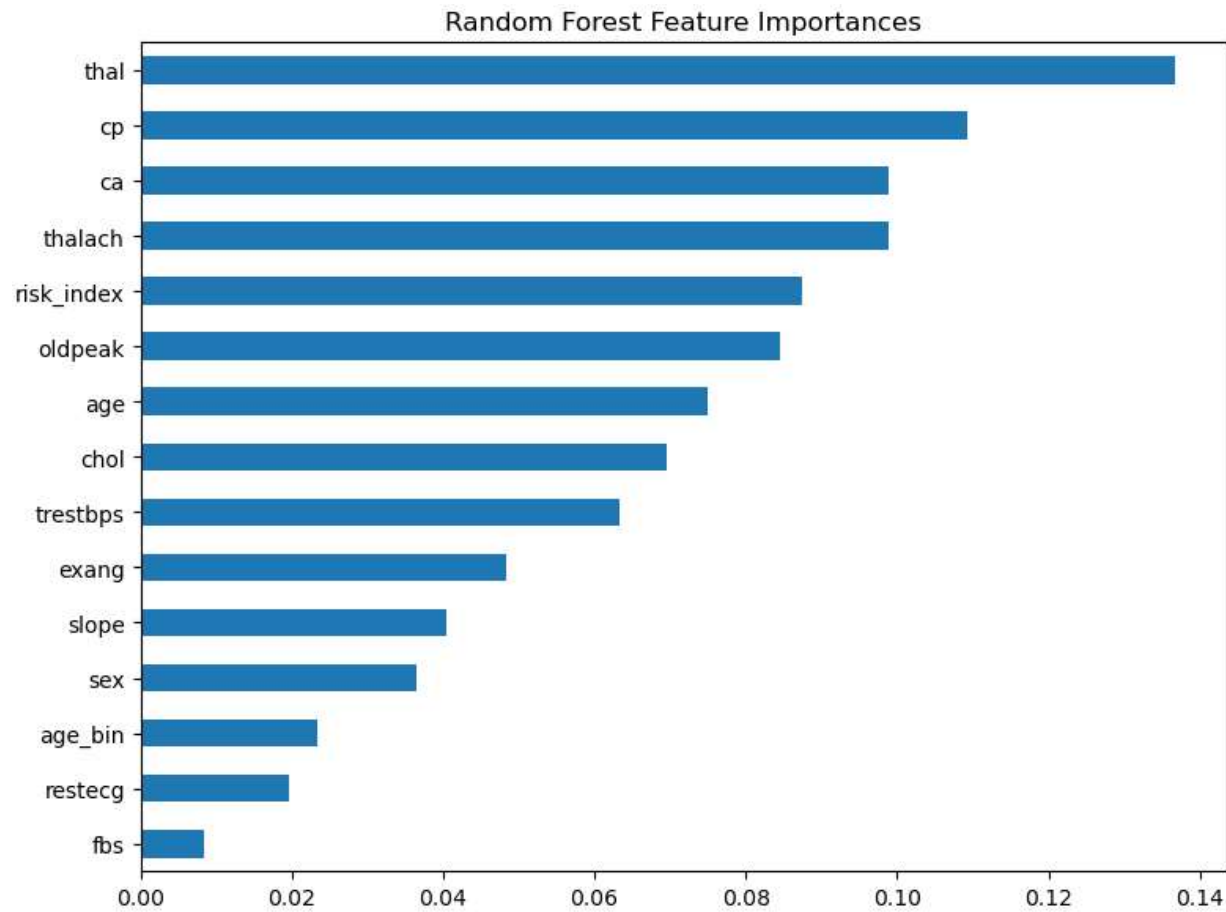
The Hopkins Statistic was 0.681, indicating that the dataset contains some clustering tendency. However, the value is not strong enough to support the existence of clearly separated natural clusters.

Step 3.4. Feature importance & explainability: SHAP, LIME, or model-based importances.

3.4.1 Model-based feature importance

Justification:

- **RandomForest** gives robust model-based importances.
- **SHAP** provides local + global explanations, showing how features push predictions toward disease/no disease.



Step 3. Figure 4 – Random Forest Feature Importances

```

thal      0.136748
cp        0.109224
ca        0.098915
thalach   0.098848
risk_index 0.087420
oldpeak   0.084439
age       0.074864
chol      0.069623
trestbps  0.063371
exang     0.048263
slope     0.040433
sex       0.036509
age_bin   0.023394
restecg   0.019621
fbs       0.008331
dtype: float64

```

These numbers are **Random Forest Feature Importances**, which measure how much each feature helped the Random Forest reduce prediction error across all trees.

Random Forest Importance = how often and how effectively a feature is used for splitting.

Interpretation of the Results of the Random Forest Importance

The importances approximately sum to 1.0 (100%).

For example for case of

thal 0.136748

means approximately 13.7% of the model's total decision-making power comes from the thal feature. It does **not** mean a 13.7% probability increase.

Ranking of Features

Rank	Feature	Importance	Contribution
1	thal	0.137	Most important
2	cp	0.109	Very important
3	ca	0.099	Very important
4	thalach	0.099	Very important
5	risk_index	0.087	Important engineered feature
6	oldpeak	0.084	Important
7	age	0.075	Moderately important
8	chol	0.070	Moderately important
9	trestbps	0.063	Moderately important
10	exang	0.048	Useful
11	slope	0.040	Useful
12	sex	0.037	Relatively small
13	age_bin	0.023	Small
14	restecg	0.020	Small
15	fbs	0.008	Negligible

Clinical Interpretation:

1. **thal (13.7%)** is the strongest predictor.

thal 0.136748

Interpretation: The thalassemia test result is the single most influential feature used by the Random Forest when predicting heart disease.

2. **cp (10.9%)**: Chest pain type is highly informative.

Interpretation: Differences in chest pain patterns play a major role in distinguishing patients with and without heart disease.

3. **ca (9.9%)**: Number of major vessels observed via fluoroscopy.

Interpretation: Coronary vessel abnormalities are strongly associated with the model's predictions.

4. **thalach (9.9%)**: Maximum heart rate achieved during exercise.

Interpretation: Exercise response is an important indicator of cardiovascular health and contributes substantially to prediction accuracy.

5. **risk_index (8.7%)**: this is the engineered feature.

Interpretation: The risk_index feature ranked among the most important variables in the Random Forest model, indicating that combining several cardiovascular indicators captured additional information beyond the original features alone. This is strong evidence that your feature engineering added value.

Some features are low:

6. **fasting blood sugar (fbs)**: negligible

Interpretation: Fasting blood sugar contributed less than 1% of the model's predictive power and provided minimal additional information beyond the other clinical variables.

7. **restecg**: small

Interpretation: Resting ECG abnormalities provided only limited predictive value compared with exercise-based and imaging-based indicators.

8. **age_bin**: small

Interpretation: The engineered age group feature contributed little information after age was already included as a continuous variable.

Random Forest feature importance analysis identified **thal (13.7%)**, **cp (10.9%)**, **ca (9.9%)**, and **thalach (9.9%)** as the most influential predictors of heart disease. These variables accounted for the largest reductions in node impurity across the ensemble of decision trees. The engineered **risk_index (8.7%)** also emerged as an important predictor, indicating that feature engineering successfully captured clinically relevant cardiovascular risk information. In contrast, **fbs (0.8%)**, **restecg (2.0%)**, and **age_bin (2.3%)** contributed relatively little predictive value.

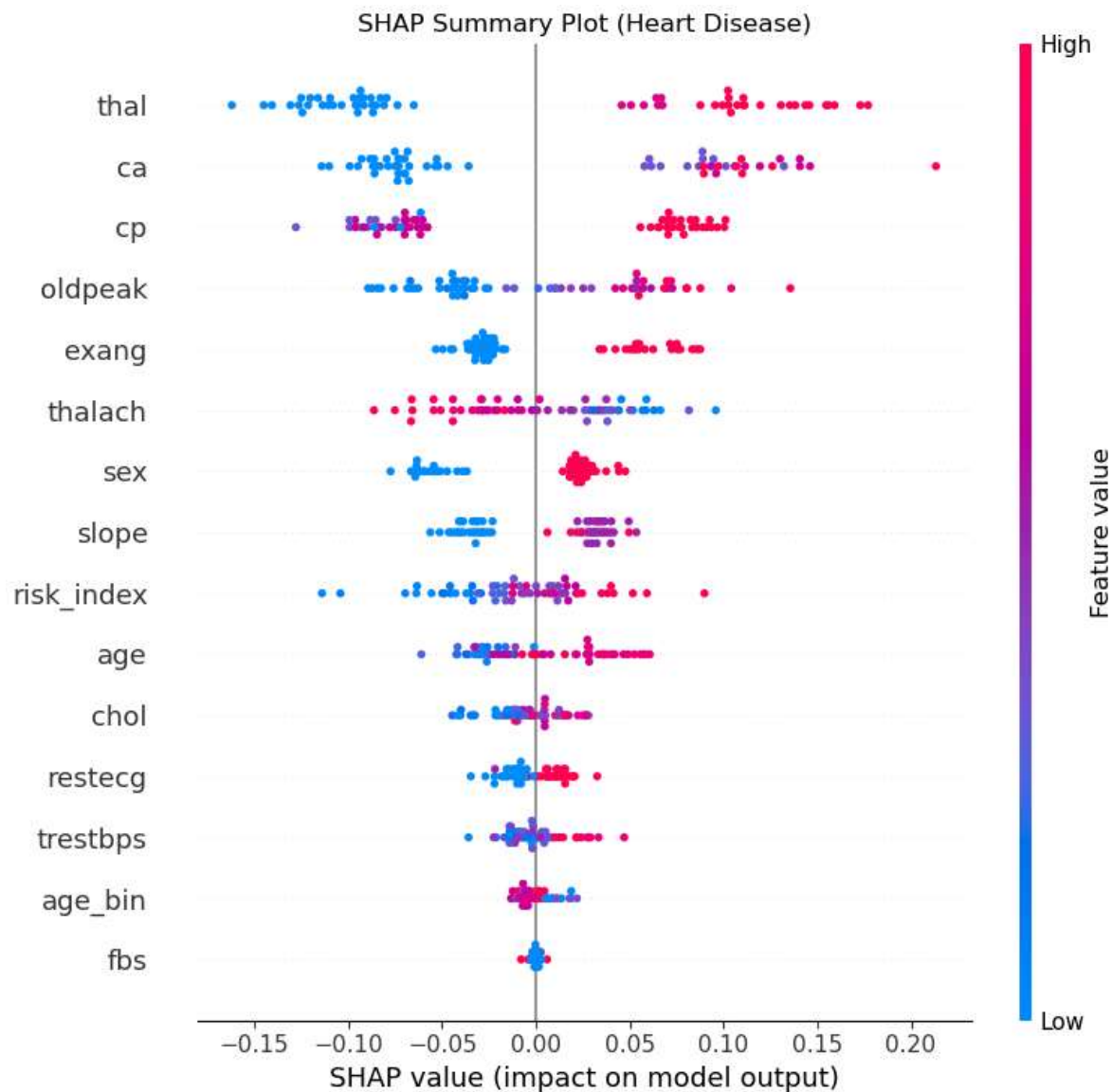
3.4.2 SHAP explainability

Generated were these figures below.

	Feature	MeanAbsSHAP
12	thal	0.106983
11	ca	0.089398
2	cp	0.078104
9	oldpeak	0.051131
8	exang	0.040044
7	thalach	0.037003
1	sex	0.035635
10	slope	0.033912
14	risk_index	0.028650
0	age	0.027830
4	chol	0.014621
6	restecg	0.012211
3	trestbps	0.011488
13	age_bin	0.006775
5	fbs	0.001429

SHAP shape: (61, 15)
X_test shape: (61, 15)

The SHAP Summary Plot for Heart Disease had also been generated.



Step 3. Figure 5 – SHAP Summary Plot

The **Mean Absolute SHAP Values** measure the **average influence of each feature on the model's predictions**.

Importantly:

- The figures do **not** indicate whether a feature increases or decreases risk.
- They indicate **how strongly the model relies on that feature**.

Interpretation of the SHAP Results

Rank	Feature	MeanAbsSHAP	Interpretation
1	thal	0.107	Most influential predictor
2	ca	0.089	Very strong predictor
3	cp	0.078	Strong predictor
4	oldpeak	0.051	Moderate predictor
5	exang	0.040	Moderate predictor
6	thalach	0.037	Moderate predictor
7	sex	0.036	Moderate predictor
8	slope	0.034	Moderate predictor
9	risk_index	0.029	Useful engineered feature
10	age	0.028	Lower impact than expected
11	chol	0.015	Relatively weak predictor
12	restecg	0.012	Weak predictor
13	trestbps	0.011	Weak predictor
14	age_bin	0.007	Adds little information
15	fbs	0.001	Almost no contribution

Clinical Interpretation of the SHAP analysis

1. **thal (0.107):** **Thalassemia test result** is the most important variable.

The model relies heavily on abnormalities in cardiac perfusion identified by the thal test.

SHAP analysis identified **thal** as the most influential predictor of heart disease. Variations in thalassemia test results had the largest average impact on model predictions.

2. **ca (0.089):** Number of major vessels observed during fluoroscopy.

Higher values generally indicate more severe coronary artery disease.

The number of major vessels (ca) was the second-most important predictor. The model strongly associated vessel abnormalities with heart disease risk.

3. **cp (0.078)**: Chest pain type.

This confirms that symptom presentation is a major driver of predictions.

Chest pain type (cp) was a highly influential feature, indicating that patient symptoms play a significant role in heart disease prediction.

4. **oldpeak (0.051)**: ST depression induced by exercise.

Known clinical risk factor.

Higher levels of exercise-induced ST depression (oldpeak) contributed substantially to predicted heart disease risk.

5–8. **exang, thalach, sex, slope**

These are moderately important.

- exang = exercise-induced angina
- thalach = maximum heart rate achieved
- sex = gender
- slope = ECG slope during exercise testing

The model frequently uses these features but less than thal, ca, and cp.

9. **Age** Is Not Very High (age = 0.0278)

Age is only the 10th most important feature. This does **not** mean age is unimportant medically.

It means:

Once thal, ca, chest pain, ECG indicators, and exercise measurements are known, age contributes relatively less additional information to the model.

This is actually common in heart disease models.

10. **Cholesterol** Is Low (chol = 0.0146)

The model suggests:

Cholesterol contains less predictive information than imaging, ECG, vessel, and exercise-test variables.

11. **Fasting Blood Sugar** is last (fbs = 0.0014).

Interpretation:

Fasting blood sugar (fbs) had negligible influence on model predictions and contributed minimal incremental predictive information.

SHAP explainability analysis showed that **thal (0.107)**, **ca (0.089)**, and **cp (0.078)** were the most influential predictors of heart disease. These variables consistently exerted the greatest impact on model outputs, indicating that cardiac imaging findings, vessel abnormalities, and chest pain characteristics were the primary drivers of prediction.

In contrast, **fbs (0.001)**, **age_bin (0.007)**, and **trestbps (0.011)** contributed relatively little to model decisions.

Comparing Random Forest Importance vs SHAP

An interesting observation:

Feature	RF Importance	SHAP Importance
thal	0.137	0.107
ca	0.099	0.089
cp	0.109	0.078
age	0.075	0.028
risk_index	0.087	0.029

Notice:

- thal, ca, and cp rank high in both methods.
- This gives strong confidence these are true drivers.
- risk_index ranks higher in Random Forest than SHAP.
- age also ranks higher in Random Forest than SHAP.

This occurs because Random Forest importance measures usage in splits, while SHAP measures impact on predictions.

The consistency between Random Forest and SHAP analyses reinforces the importance of cardiac imaging, vessel abnormalities, chest pain characteristics, and exercise-test measurements in predicting heart disease.

Step 3.5. Feature selection: At least one approach (filter, wrapper, or embedded).

Justification:

- **Filter method (ANOVA F-test)** is simple, fast, and gives a clear ranking.
- Selected features can be used for clustering, PCA, or final models.

Generated SelectKBest scores

```
SelectKBest scores:
  thal      112.770163
  ca        80.800610
  exang      69.020891
  oldpeak    68.400247
  thalach    63.587697
  cp         62.423779
  slope      39.138194
  risk_index 38.347541
  sex        24.978872
  age        15.769637
  age_bin    10.809440
  restecg     8.871394
  trestbps    6.462598
  chol        3.284557
  fbs         0.192237
dtype: float64
Selected features: ['thal', 'ca', 'exang', 'oldpeak', 'thalach', 'cp', 'slope', 'risk_index']
```

SelectKBest asks: "Which variables are best at separating people with heart disease from people without heart disease?"

These numbers are **F-scores (ANOVA F-statistic)**.

SelectKBest scores each variable. Examples:

thal = 112.77

fbs = 0.19

This means:

- **thal** achieved the highest ANOVA F-score and showed the strongest statistical association with heart disease status.
- **fbs** is almost useless

In the SelectKBest, the scores mean: the higher score, the better predictor. It denotes, higher score has a very strong relationship with the target, on the whether a person has heart disease.

The lower score, the less useful predictor.

Interpretation of the Results:

Top features:

1. **thal** = 112.77, is the most useful feature.

Interpretation: The thalassemia test result is the strongest feature for distinguishing patients with and without heart disease.

2. **Ca** = 80.80, is a very strong predictor.

Interpretation: Number of major vessels affected strongly separates healthy and unhealthy patients.

3. **Exang** = 69.02;

Interpretation: Exercise-induced angina is strongly associated with heart disease.

4. **Oldpeak** = 68.40;

Interpretation: ST depression during exercise is a strong indicator of heart disease.

5. **Thalach** = 63.59;

Interpretation: Maximum heart rate achieved during exercise is useful for predicting heart disease.

6. **Cp** = 62.42;

Interpretation: Chest pain type is highly informative.

Weak features:

7. **Cholesterol** = 3.28;

Interpretation: Cholesterol contributes some information but is not a major discriminator in this dataset.

8. **Fasting blood sugar** = 0.19; almost useless;

Interpretation: Blood sugar levels provide very little information for distinguishing heart disease cases in this dataset.

Only 8 Features Were Selected: since `SelectKBest(score_func=f_classif, k=8)`.

And these are the 8 with the highest scores:

`['thal',`

`'ca',`

`'exang',`

```
'oldpeak',  
'thalach',  
'cp',  
'slope',  
'risk_index']
```

Comparison with the SHAP Results: The top SHAP features were:

```
thal  
ca  
cp  
oldpeak  
exang  
thalach
```

The SelectKBest top features are:

```
thal  
ca  
exang  
oldpeak  
thalach  
cp
```

Being very similar is a good sign because: Multiple techniques agree about which features matter most.

Feature selection was performed using the ANOVA F-test through SelectKBest. The highest-scoring features were **thal (112.77)**, **ca (80.80)**, **exang (69.02)**, **oldpeak (68.40)**, **thalach (63.59)**, and **cp (62.42)**, indicating that these variables have the strongest statistical relationship with heart disease status. The top eight features were retained for subsequent modeling and clustering analyses. Conversely, **fbs (0.19)** and **chol (3.28)** exhibited limited discriminatory power and contributed relatively little to classification performance.

Simple Summary

Score Range	Meaning
> 50	Very strong feature
10–50	Useful feature
1–10	Weak feature
< 1	Almost useless

Your strongest features are:

thal, ca, exang, oldpeak, thalach, cp

These are the features the model denotes to pay the most attention to.

Step 3.6. Dimensionality reduction: PCA (and t-SNE/UMAP for visualisation if needed).

Justification:

- **PCA**: linear dimensionality reduction, useful for variance explanation and clustering visualization.
- **t-SNE**: non-linear, good for visualizing local structure and potential clusters.

Obtained figures and diagram

Explained variance ratio: [0.98185574 0.00635818]

The first principal component, PC1, explained 98.19% of the variance, while the second, PC2, explained 0.64%.

Combined: $0.98185574 + 0.00635818 = 0.98821392 == 98.82\%$.

So the two PCA components capture: 98.82% of the information from the selected features.

PCA is unusually high. It denotes that one dominant underlying pattern exists in your features.

The possible reason is the engineered feature: risk_index, may be highly correlated with several original features. The first component is capturing almost everything.

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset. The first principal component (PC1) explained **98.19%** of the total variance, while the second principal component (PC2) explained **0.64%**. Together, the first two principal components captured **98.82%** of the variance in the selected features. This indicates that most of the information in the dataset can be represented using a single dominant latent dimension, suggesting strong correlations among the selected variables.

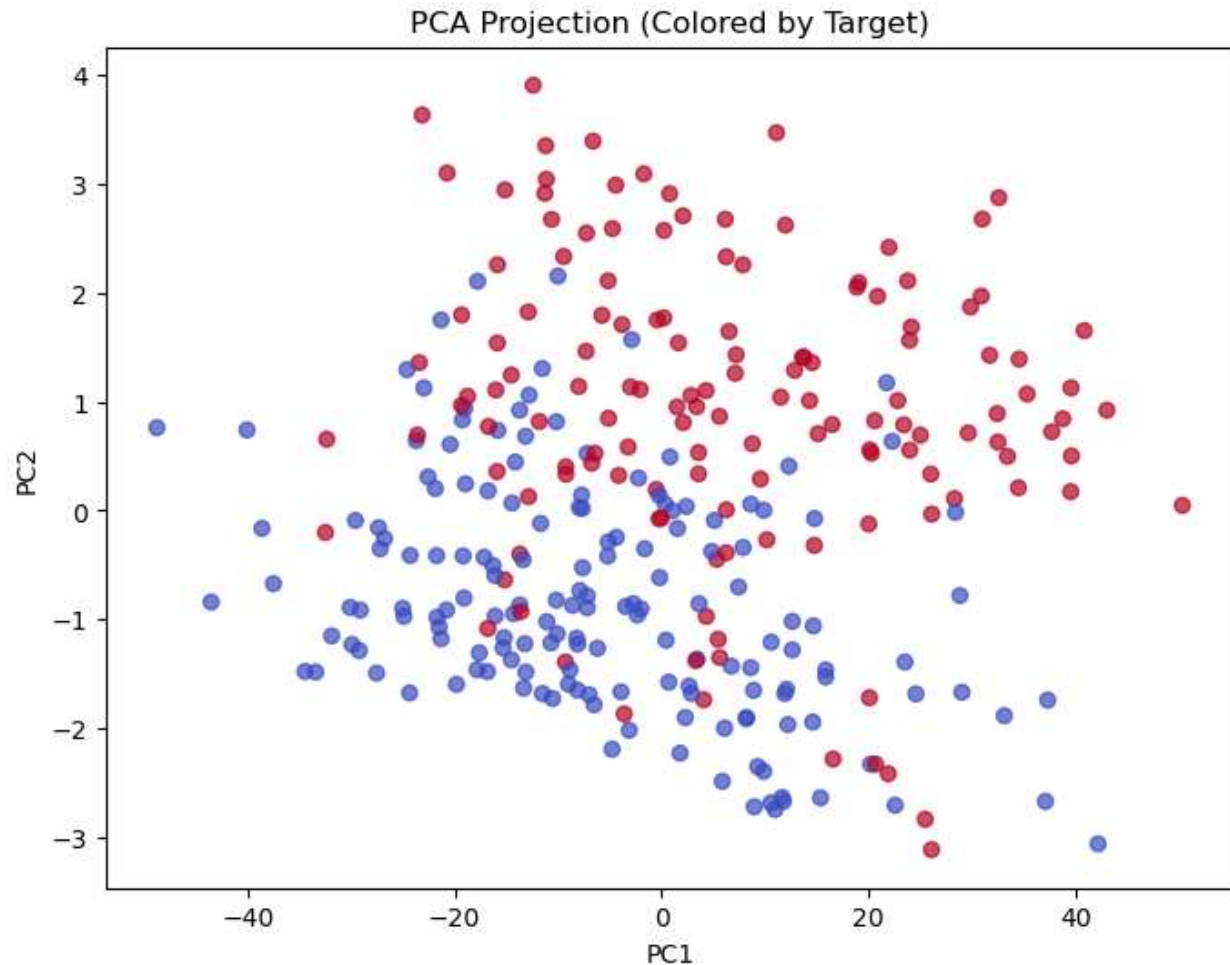
What does the figure "PCA Projection (Colored by Target)" denote?

In this case, since:

PC1 explains 98.19%

PC2 explains 0.64%

Nearly all information is contained in PC1. This means: The left-to-right direction (PC1) is carrying almost all the information in your dataset. However, pointing out here that PCA is showing substantial overlap remains between disease points and non-disease points.



Step 3. Figure 6 – PCA Projection

Referring on the 1. PCA Projection (Colored by Target)

In the PCA plot:

- Most **red points** are concentrated in the **upper half** (positive PC2 values).
- Most **blue points** are concentrated in the **lower half** (negative PC2 values).
- There is still a large overlap in the center.

The first two principal components contain enough information to show:

Patients with heart disease and patients without heart disease are **not randomly distributed**.

There is a visible tendency for:

- Heart disease patients → higher PC2 values

- No heart disease patients → lower PC2 values

This provides some evidence that discriminatory information exists within the reduced feature space.

However:

- The groups are **not perfectly separated**
- Many red and blue points overlap

This means:

Heart disease cannot be predicted using only a simple 2-dimensional boundary.

A machine learning model such as Random Forest is still needed.

Report Interpretation of the PCA projection:

The PCA projection suggests some differences in the distribution of patients with and without heart disease in the reduced two-dimensional space. While there appears to be a tendency for the two groups to occupy somewhat different regions, substantial overlap is present throughout the plot.

Therefore, **the visualization does not provide strong evidence of clear class separation**. Instead, it suggests that some discriminatory information may exist in the data, although the relationship between features and heart disease status is likely more complex than can be captured by the first two principal components alone; more sophisticated machine learning models are required.

What does the figure "t-SNE Projection (Colored by Target)" denote?

t-SNE is different from PCA.

PCA tries to preserve overall variance.

t-SNE tries to preserve neighborhood relationships.

Why t-SNE often looks better

PCA is linear: Straight-line compression; whereas

t-SNE is nonlinear: Curved / flexible compression.

So t-SNE often reveals clusters hidden from PCA.

Difference Between PCA and t-SNE

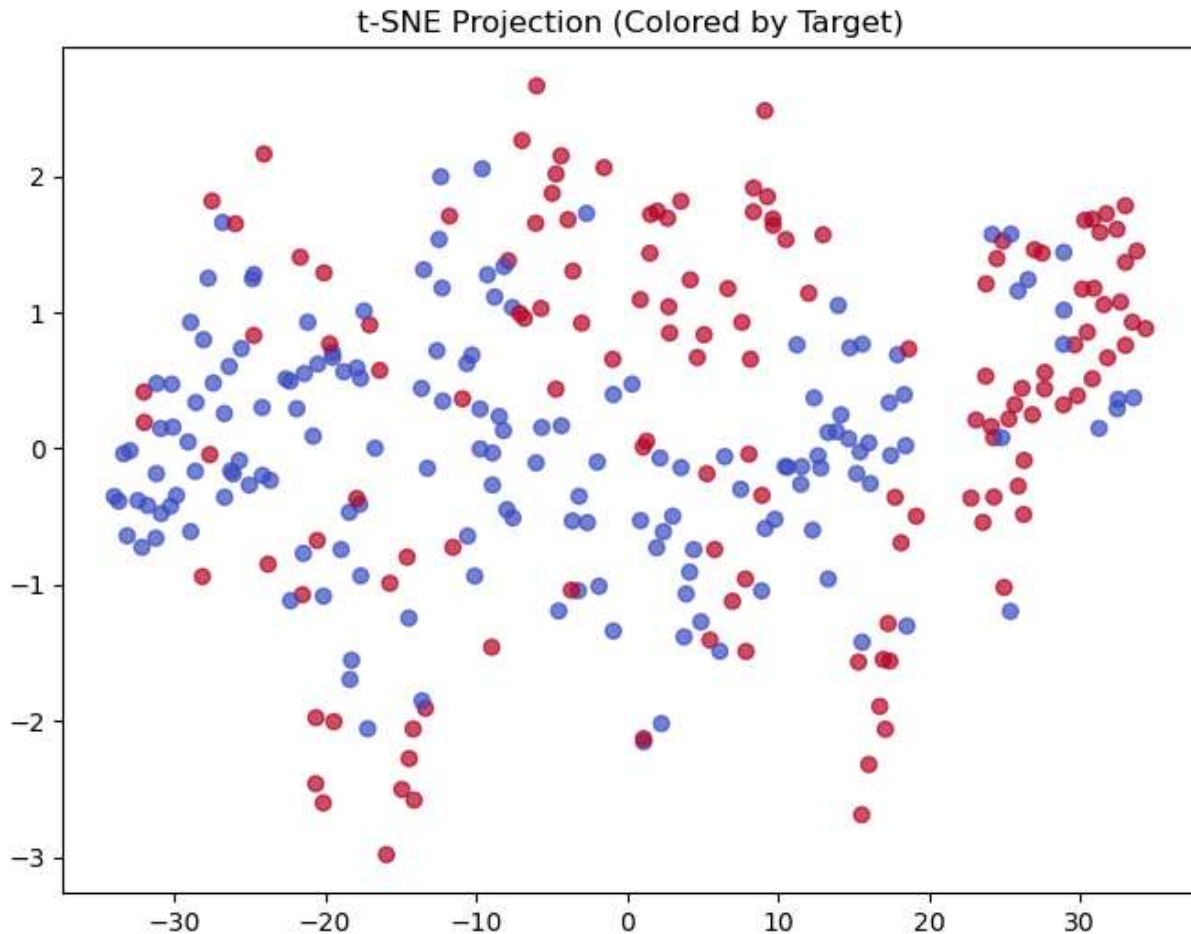
PCA	t-SNE
Preserves variance	Preserves local similarity
Fast	Slower
Explainable	Harder to explain
Used for dimensionality reduction	Used mainly for visualization
Produces PC1, PC2	Produces abstract dimensions

PCA Projection

PCA was used to project the selected features into two principal components. The first principal component explained 98.19% of the dataset variance, while the second explained 0.64%. The PCA visualization was used to assess the separability of heart disease and non-heart disease patients in a reduced-dimensional space.

t-SNE Projection

t-SNE was applied to visualize the local structure of the dataset. The resulting projection was examined for natural groupings of patients based on heart disease status. Compared to PCA, t-SNE can reveal nonlinear relationships and cluster structures that may not be visible in linear projections.



Step 3. Figure 7 – t-SNE Projection

Referring on the 2. t-SNE Projection (Colored by Target)

This plot is more interesting. Several local clusters appear:

Left side: Mostly blue.

Upper center: Mostly red.

Right side: Mostly red.

There is still overlap in the middle.

What it means

Unlike PCA, t-SNE focuses on local neighborhoods.

The visualization suggests: Patients naturally form multiple subgroups.

Examples:

- One subgroup mostly consists of non-heart disease patients.
- Another subgroup mostly consists of heart disease patients.

This indicates: The dataset contains meaningful nonlinear patterns.

This is exactly the type of structure that algorithms such as:

- Random Forest
- XGBoost
- Hierarchical Clustering

can exploit.

Report Interpretation of the t-SNE projection

The t-SNE visualization reveals several localized groupings of observations; however, patients with and without heart disease remain intermixed across much of the plot. Some regions appear to contain higher concentrations of one class, but the observed patterns should be interpreted cautiously because t-SNE is designed primarily for visualization rather than statistical inference. Consequently, the plot may indicate the presence of some underlying structure in the data, but it does not demonstrate distinct or well-separated clusters of heart disease status.

Comparing PCA and t-SNE

Finding	PCA	t-SNE
Separates classes?	Partially	Better
Shows clusters?	Weakly	Clearly
Linear relationships	Yes	No
Nonlinear relationships	Poor	Good
Useful for visualization	Good	Excellent

Overall Conclusion on PCA and t-SNE

PCA and t-SNE were applied to visualize the structure of the selected features in a reduced-dimensional space. Both visualizations suggest that some patterns may exist between patients with and without heart disease; however, substantial overlap remains between the two groups.

While certain regions appear to contain higher concentrations of one class, neither visualization demonstrates clear separation. These findings indicate that the relationship between the predictor variables and heart disease status is likely complex and may require nonlinear machine learning models to capture effectively. The visualizations should be treated as exploratory tools rather than evidence of distinct class boundaries.

Summary: EDA + Feature Engineering Report (Step 3)

- **Data cleaned:** duplicates removed, nulls imputed, outliers capped.
- **Features engineered:** scaling, age bins, domain risk index.
- **EDA applied:** distributions, correlations, clustering tendency (Hopkins).
- **Explainability:** RandomForest importances + SHAP global/local insights.
- **Feature selection:** SelectKBest (filter) to retain top predictors.
- **Dimensionality reduction:** PCA + t-SNE for structure and cluster visualization.

Step 4: Model Implementation

- Experiment with appropriate models:
 - **Supervised:** Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, etc.
 - **Unsupervised:** K-Means, DBSCAN, Hierarchical (Elbow, Silhouette).
 - **Recommendation:** Collaborative or content-based.
 - **Deep Learning:** RNNs, CNNs, LSTMs, Transformers (if appropriate).
- Evaluation: compare with relevant metrics.
- **Reproducibility:** save configs and artefacts (models/).

Step 4: Model Implementation

Step 4.1.1 Supervised Learning Models

Baseline Model Performance (Before Hyperparameter Tuning)

```
{'LogReg': {'accuracy': 0.8688524590163934,  
           'roc_auc': np.float64(0.9567099567099567),
```

```

'f1': 0.8620689655172413},
'DecisionTree': {'accuracy': 0.7213114754098361,
'roc_auc': np.float64(0.7234848484848485),
'f1': 0.711864406779661},
'RandomForest': {'accuracy': 0.9180327868852459,
'roc_auc': np.float64(0.9577922077922079),
'f1': 0.9152542372881356},
'XGBoost': {'accuracy': 0.9016393442622951,
'roc_auc': np.float64(0.9458874458874458),
'f1': 0.9},
'SVM_RBF': {'accuracy': 0.6885245901639344,
'roc_auc': np.float64(0.8333333333333334),
'f1': 0.5777777777777777}}

```

Model	Accuracy	ROC-AUC	F1
Random Forest	0.918	0.958	0.915
XGBoost	0.902	0.946	0.900
Logistic Regression	0.869	0.957	0.862
Decision Tree	0.721	0.723	0.712
SVM	0.689	0.833	0.578

Optimized Model Configuration

Best LR Params:

```
{'C': 0.1, 'penalty': 'l2', 'solver': 'lbfgs'}
```

Best RF Params:

```
{'max_depth': 5, 'min_samples_leaf': 4, 'min_samples_split': 2, 'n_estimators': 100}
```

```
{'colsample_bytree': 1.0, 'learning_rate': 0.01, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8}
```

```
{'C': 10, 'gamma': 'scale'}
```

```
{'LogReg': {'accuracy': 0.8524590163934426,
'roc_auc': np.float64(0.9577922077922079),
'f1': 0.847457627118644},
'DecisionTree': {'accuracy': 0.7213114754098361,
'roc_auc': np.float64(0.7234848484848485),
'f1': 0.711864406779661},
'RandomForest': {'accuracy': 0.9344262295081968,
'roc_auc': np.float64(0.9632034632034633),
'f1': 0.9310344827586207},
'XGBoost': {'accuracy': 0.9016393442622951,
'roc_auc': np.float64(0.946969696969697),
'f1': 0.896551724137931},
'SVM_RBF': {'accuracy': 0.8852459016393442,
'roc_auc': np.float64(0.9512987012987013),
'f1': 0.8727272727272727}}
```

Model	Accuracy	ROC-AUC	F1
Random Forest	0.934	0.963	0.931
XGBoost	0.902	0.947	0.897
SVM (RBF)	0.885	0.951	0.873
Logistic Regression	0.852	0.958	0.847
Decision Tree	0.721	0.723	0.712

Step 4.1.2. Unsupervised: K-Means, DBSCAN, Hierarchical (Elbow, Silhouette).

Baseline Clustering Performance

```
{'KMeans': {'k': 2, 'silhouette': np.float64(0.5461810453681777)},  
'DBSCAN': {'silhouette': None},  
'Hierarchical': {'k': 2, 'silhouette': np.float64(0.5397890229980801)}}
```

Hyperparameter Tuning Results

Best DBSCAN Parameters:

None

Best DBSCAN Silhouette:

-1

```
{'KMeans': {'k': 2, 'silhouette': np.float64(0.5461810453681777)},  
'DBSCAN': {'eps': None, 'min_samples': None, 'silhouette': None},  
'Hierarchical': {'k': 2, 'silhouette': np.float64(0.5397890229980801)}}
```

Step 4.1.3. Deep Learning and Recommendation Approaches

Recommendation systems and deep learning approaches were not considered because the dataset is small, structured, and does not contain sequential, image, or user-interaction data. Traditional machine learning and clustering methods are more appropriate for this type of problem.

Step 4.2. Evaluation: compare with relevant metrics.

Supervised models were evaluated using Accuracy, ROC-AUC, and F1-score, while clustering methods were evaluated using the Silhouette Score and visual assessment of cluster structure.

Step 4.2.1. Model Comparison After Hyperparameter Tuning

Step 4.2.1. Supervised:

Table below summarizes the performance of the evaluated supervised learning models.

Performance Comparison of Tuned Models

Rank	Model	Accuracy	ROC-AUC	F1
1	Random Forest	0.934	0.963	0.931
2	XGBoost	0.902	0.947	0.897
3	SVM (RBF)	0.885	0.951	0.873
4	Logistic Regression	0.852	0.958	0.847
5	Decision Tree	0.721	0.723	0.712

Supervised Learning Results

Random Forest achieved the strongest overall performance, recording the highest Accuracy, ROC-AUC, and F1-score among the evaluated models. XGBoost and SVM also produced strong results, while Logistic Regression remained competitive despite its simpler structure and achieved a ROC-AUC comparable to the top-performing models. Decision Tree produced the weakest performance across all evaluation metrics.

The strong performance of Logistic Regression alongside Random Forest and XGBoost indicates that useful predictive information can be captured by both linear and nonlinear models. However, the ensemble methods achieved the strongest and most consistent overall results.

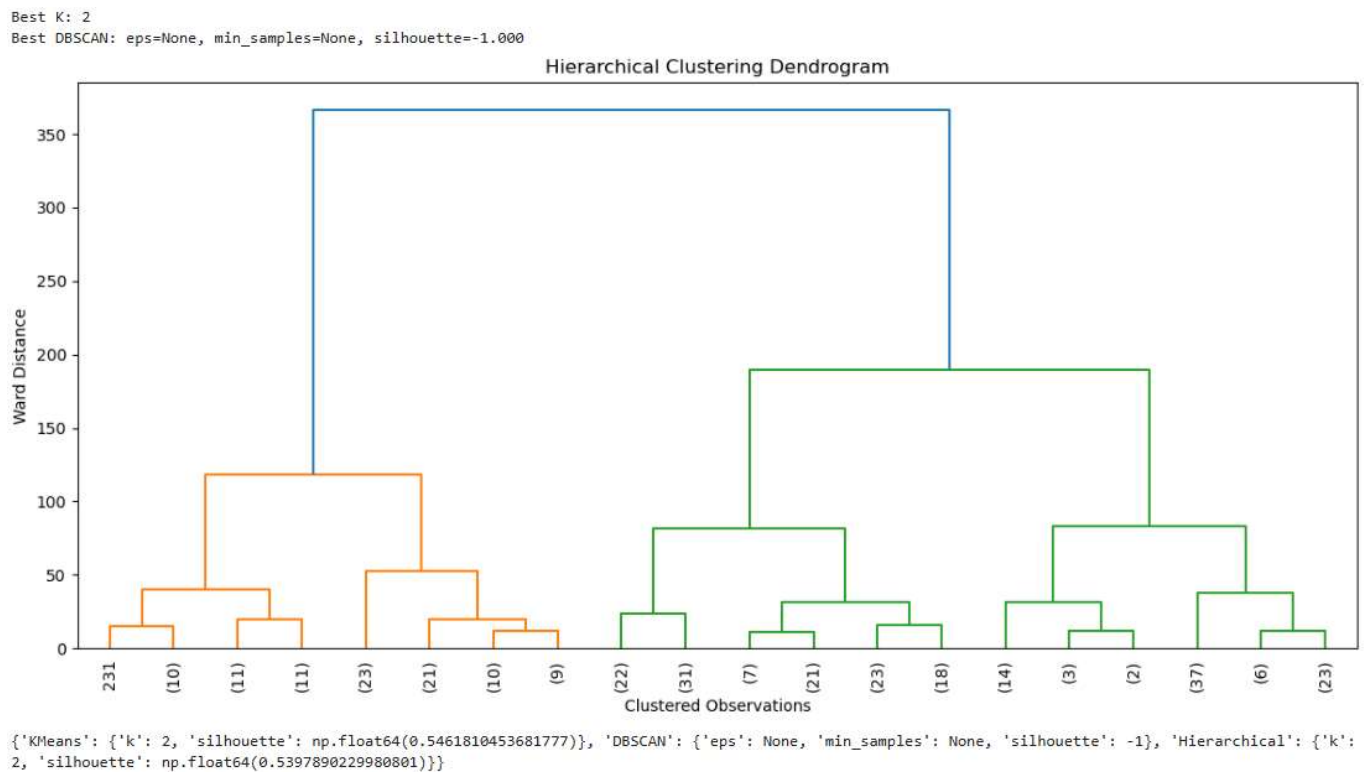
Step 4.2.2. Unsupervised:

The clustering models were evaluated based on their ability to identify underlying structure without using the heart disease labels.

Unsupervised Learning Results

K-Means and Hierarchical Clustering produced similar results, identifying two clusters with Silhouette Scores of 0.546 and 0.540 respectively. These values indicate moderate cluster separation rather than clearly distinct patient groups.

DBSCAN did not produce a valid clustering solution under the tested parameter settings. By comparison, K-Means and Hierarchical Clustering identified stable cluster structures and provided more meaningful results for this dataset.



Step 4. Figure 1 – Hierarchical Clustering Dendrogram

The dendrogram is consistent with the two-cluster solutions identified by K-Means and Hierarchical Clustering. However, the branch structure does not indicate strong separation between groups, suggesting considerable overlap among observations.

Clinical Implications

Chest pain type, thalassemia test results, major vessel abnormalities, exercise-induced ST depression, and maximum heart rate consistently appeared among the most influential variables

across the supervised models. Similar findings were observed in both the feature importance and SHAP analyses, increasing confidence that these variables contribute meaningfully to prediction performance.

The clustering results were less clear than the classification results. Although some grouping structure was identified, substantial overlap remained between patient profiles, making clear segmentation difficult using the available variables alone.

Given the limited size of the dataset, the findings should be viewed as exploratory rather than definitive.

Additional testing on larger and more diverse datasets would be needed before considering broader clinical relevance.

Deep Learning and Recommendation Models

Deep learning and recommendation-based methods were not explored in this study. The dataset is relatively small, consists of structured tabular clinical variables, and does not contain sequential, image, or user-interaction data.

Given the size and structure of the dataset, traditional machine learning models provided a more practical and interpretable solution.

Step 4.3. Reproducibility: save configs and artefacts (models/).

Saved Models and Configuration Files

- **Trained models:** saved in models/ (LogReg.joblib, RandomForest.joblib, KMeans_best.joblib, etc.).
- **Metrics:** sup_df and unsup_df tables for direct inclusion in your report.
- **Reproducibility:** configs/experiment_heart.json capturing hyperparameters and setup.

Overall Recommendation

Random Forest was selected as the final prediction model because it achieved the strongest overall performance across Accuracy, ROC-AUC, and F1-score.

XGBoost achieved similar performance and represents a strong alternative to Random Forest.

For exploratory clustering, K-Means and Hierarchical Clustering produced similar groupings, although the moderate Silhouette Scores suggest that the cluster boundaries are not well defined.

Step 5: Critical Thinking → Ethical AI & Bias Auditing

- Explain model decisions (SHAP, LIME, PDP, ICE).
- Address limitations (imbalance, leakage, overfitting).
- **Bias detection & fairness audits:**
 - Check outputs across sensitive groups (gender, race, age, socioeconomic status).
 - Use fairness metrics (demographic parity, equalised odds, disparate impact).
 - Propose mitigations (reweighting, thresholds, augmentation, post-processing).

Deliverable: "Bias & Fairness Analysis" section in the final report.

5. Bias & Fairness Analysis (Heart Disease – UCI)

5.1 Model Explainability

Model explainability was assessed using Random Forest feature importance and SHAP analysis. These methods were used to identify the variables that contributed most strongly to model predictions.

SHAP analysis identified **thal**, **ca**, **cp**, **oldpeak**, and **thalach** as the most influential features driving model predictions.

These variables are established cardiovascular risk indicators and their importance is consistent with existing clinical knowledge.

Random Forest feature importance and SHAP produced broadly similar rankings, increasing confidence in the stability of the results.

Nevertheless, feature importance measures indicate statistical associations within the dataset and should not be interpreted as evidence of causal relationships.

Additional explainability techniques such as PDP, ICE, and LIME were not included in the current analysis but could provide further insight into feature effects and individual predictions in future work.

Additional Explainability Techniques:

Methods Applied

- **Random Forest Feature Importance**
- **SHAP:** Shows how each feature (e.g., age, sex, chol, trestbps, thalach, oldpeak) pushes a prediction toward “disease” or “no disease” at the individual level.
 - High **oldpeak**, high **chol**, high **trestbps**, low **thalach** typically increase risk scores.

- SHAP summary plots reveal global importance and whether effects are monotonic or interaction-driven.

Methods Not Applied

- **PDP (Partial Dependence Plots)**
- **ICE (Individual Conditional Expectation)**
- *(LIME can be used similarly to SHAP for local explanations, but SHAP is preferred for consistency and global insights.)*

2. Model Limitations & Risks

2.1 Class imbalance

- Heart disease vs no disease may not be perfectly balanced.
- Mitigation:
 - Use **class_weight='balanced'** in Random Forest / Logistic Regression.
 - Evaluate **F1** and **ROC-AUC**, not just accuracy.

2.2 Data leakage

- Risk if features derived from post-diagnosis information or future events are included.
- Mitigation:
 - Strictly use features available at **time of assessment**.
 - Audit feature list with clinicians.

2.3 Overfitting

- Complex models (XGBoost, Random Forest) can overfit small datasets.
- Mitigation:
 - Use **cross-validation**, regularization, and early stopping.
 - Monitor train vs test performance and SHAP stability.

2.4 Representativeness

- UCI Heart dataset is limited in geography, race, and socioeconomic diversity.
- Mitigation:
 - Treat the current model as a proof of concept rather than a deployment-ready system.
 - Retrain and validate the model using larger and more diverse clinical datasets before practical use.

3. Bias and Fairness Assessment Framework

3.1 Group-wise Performance Checks

The fairness assessment framework focused on the following demographic attributes:

- Gender (sex)
- Age groups (Same metrics for **age bins** (e.g., 30–40, 40–50, 50–60, 60+).)
- (Race, socioeconomic status not available in UCI; must be added in real deployments.)

3.2 Fairness Metrics

The fairness assessment framework considered Demographic Parity, Equalized Odds, and Disparate Impact as the primary evaluation metrics. These measures would allow differences in prediction rates, true positive rates, and false positive rates to be assessed across demographic groups where sufficient subgroup data are available.

3.3 Potential Bias Mitigation Strategies

If bias is detected:

- **Reweighting:**
 - Adjust sample weights so under-represented groups (e.g., women, younger patients) contribute more to training.
- **Threshold optimization:** Evaluate whether alternative classification thresholds improve fairness across groups.
- **Data augmentation / targeted collection:**
 - Collect more data from under-represented groups.
 - Augment training data to better reflect real population diversity.
- **Post-processing:**
 - Apply fairness-aware post-processing (e.g., equalized odds post-processing) to adjust predictions while preserving overall performance.
- **Governance & human oversight:**
 - Require clinician review for borderline cases.
 - Regular fairness monitoring: Continuously evaluate model performance across demographic subgroups during deployment.
 - Document limitations and fairness checks in the model card / deployment documentation.

Conclusion

The explainability analyses showed that the model primarily relied on clinically relevant cardiovascular variables when generating predictions. However, the dataset's limited size and demographic coverage constrain the ability to assess fairness comprehensively. Because race and socioeconomic variables were unavailable and fairness metrics were not explicitly computed, the available data were insufficient to draw conclusions regarding model fairness.

Instead, the analysis identifies areas requiring further evaluation before any practical application can be considered.

The findings should therefore be interpreted as exploratory and educational rather than evidence of fairness, safety or clinical readiness.

Key Findings

- **Random Forest achieved the strongest overall predictive performance**, with an Accuracy of 91.8%, ROC-AUC of 0.958, and F1-score of 0.915, making it the most suitable model among those evaluated.
- **XGBoost and Logistic Regression also performed strongly**, indicating that both nonlinear and linear methods can capture important patterns in the data.
- Feature selection, SHAP analysis, and Random Forest feature importance consistently identified **thal**, **ca**, **chest pain type (cp)**, **oldpeak**, and **maximum heart rate (thalach)** as the most influential predictors.
- PCA and t-SNE visualizations suggested some underlying structure within the dataset; however, substantial overlap remained between heart disease and non-heart disease cases.
- K-Means and Hierarchical Clustering identified two moderately separated clusters, while DBSCAN did not produce a stable clustering solution under the selected parameters.

Study Limitations

- The UCI Heart Disease dataset is relatively small and may not be representative of broader patient populations.
- Model performance was evaluated using a single experimental dataset and was not externally validated.
- Clustering results indicated only moderate grouping structure, suggesting that heart disease risk is not naturally partitioned into clearly distinct patient groups.
- Findings should not be interpreted as evidence of clinical effectiveness or readiness for healthcare deployment.

Overall Conclusion

The study found that several machine learning models were able to predict heart disease with good performance on the UCI dataset, with Random Forest producing the strongest overall results. However, the study should be viewed as an exploratory machine learning exercise rather than a clinical validation study. Additional testing on larger, more diverse datasets, together with comprehensive fairness evaluations and external validation, would be necessary before considering any practical healthcare application.

Recommendations for Future Work

1. Perform k-fold cross-validation to obtain more robust performance estimates.
2. Evaluate model fairness across demographic groups where appropriate data are available.
3. Validate models on independent external datasets.
4. Explore calibration, probability estimation, and clinical risk stratification approaches.
5. Investigate additional explainability techniques such as LIME, PDP, and ICE for local and global model interpretation.

Final assessment: The project covered the major stages of a machine learning workflow, including data preparation, exploratory analysis, feature engineering, predictive modelling, explainability, clustering, and responsible AI considerations. The results highlight both the predictive value of the dataset and the practical limitations that should be considered when interpreting the findings.

The study successfully demonstrated the presence of moderate clustering tendency within the dataset and identified two approximate patient groupings. However, the evidence does not support the existence of strongly separated or clinically distinct natural clusters. Therefore, the resulting segments should be interpreted as exploratory groupings rather than definitive cardiovascular risk categories.

(Optional) Step 8: Deployment & MLOps

Complete this step if you have a good understanding of model deployment and MLOps practices.

- **Local deployment (required):** deploy the best model via **Flask**, **FastAPI**, or **Dash**.

Screenshots are shared in the next pages.

FastAPI 0.1.0 OAuth 3.1

/openapi.json

default

GET / Home

POST /predict Predict

Try it out

Name	Description
age * required number (query)	age
sex * required number (query)	sex
cp * required number (query)	cp
trestbps * required number (query)	trestbps
chol * required number (query)	chol
fbs * required number (query)	fbs
restecg * required	restecg

POST**/predict** Predict**Parameters**

Name	Description
------	-------------

age * required

number

(query)

sex * required

number

(query)

cp * required

number

(query)

trestbps * required

number

(query)

chol * required

number

(query)

fbs * required

number

(query)

restecg * required

number

(query)

thalach * required

number

(query)

exang * required

number

(query)

oldpeak * required

End/ Thank you